

Data Handling & Visualization

Lab Experiments

DSA0602 – SLOT A by Dr. Almas Begum

Name: Rishwanth. L

Reg. No.: 192424100

SET 21 — Energy Consumption Analysis — Experiment 1

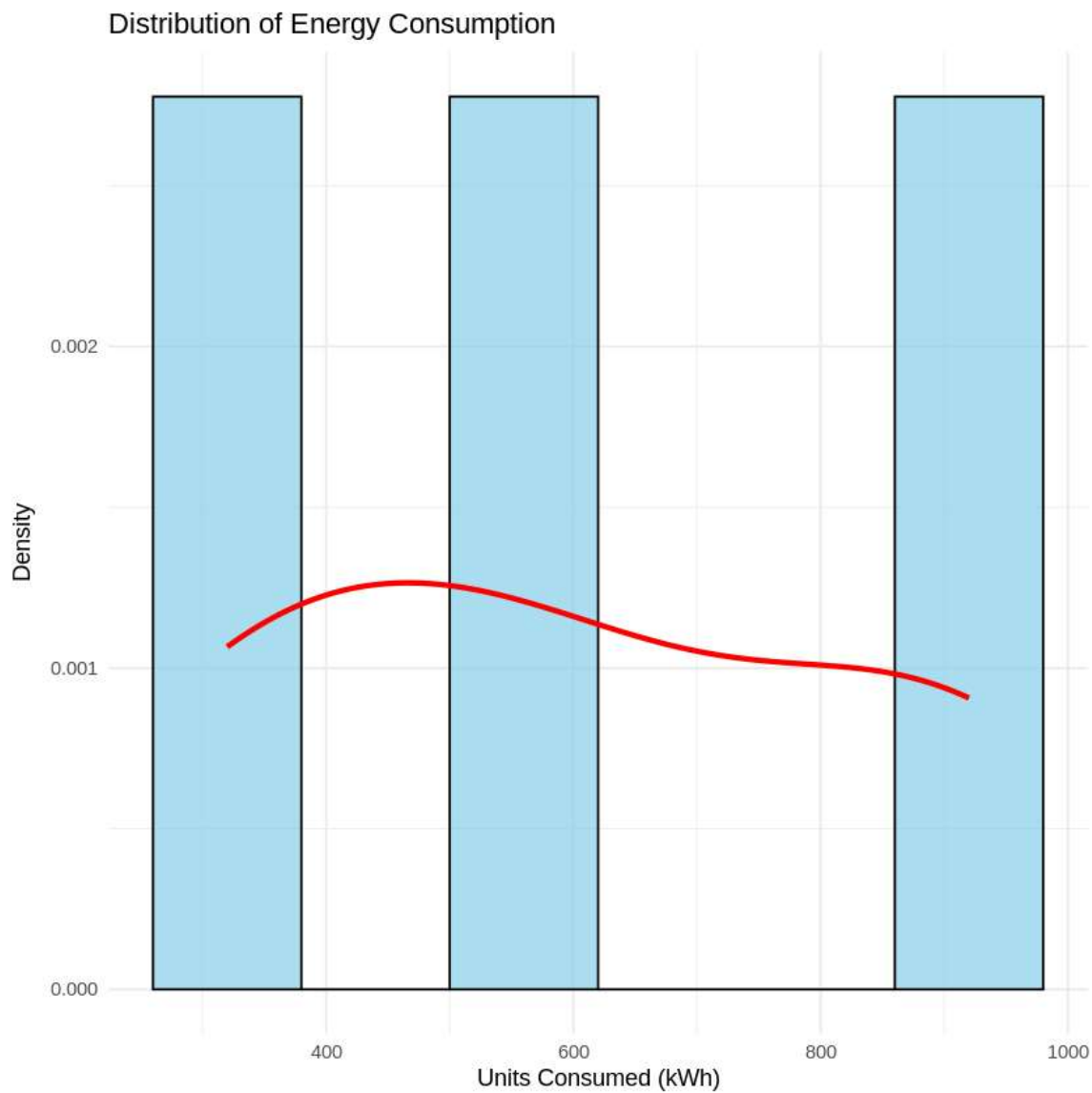
Import the dataset in R. Plot a histogram and density plot for Units_Consumed. Compare consumption patterns.

Code:

```
# Create the dataset
energy_data <- data.frame(
  Sector = c("Residential", "Commercial", "Industrial",
            "Residential", "Commercial", "Industrial"),
  Region = c("North", "South", "West", "East", "North", "South"),
  Month = c("Jan", "Jan", "Feb", "Feb", "Mar", "Mar"),
  Units_Consumed = c(320, 540, 880, 350, 610, 920),
  Peak_Hours = c(4, 6, 8, 5, 7, 9)
)

library(ggplot2)
ggplot(energy_data, aes(x = Units_Consumed)) +
  geom_histogram(
    aes(y = after_stat(density)),
    bins = 6,
    fill = "skyblue",
    color = "black",
    alpha = 0.7
  ) +
  geom_density(
    color = "red",
    linewidth = 1.2
  ) +
  labs(
    title = "Distribution of Energy Consumption",
    x = "Units Consumed (kWh)",
    y = "Density"
  ) +
  theme_minimal()
```

Output:



SET 21 — Energy Consumption Analysis — Experiment 2

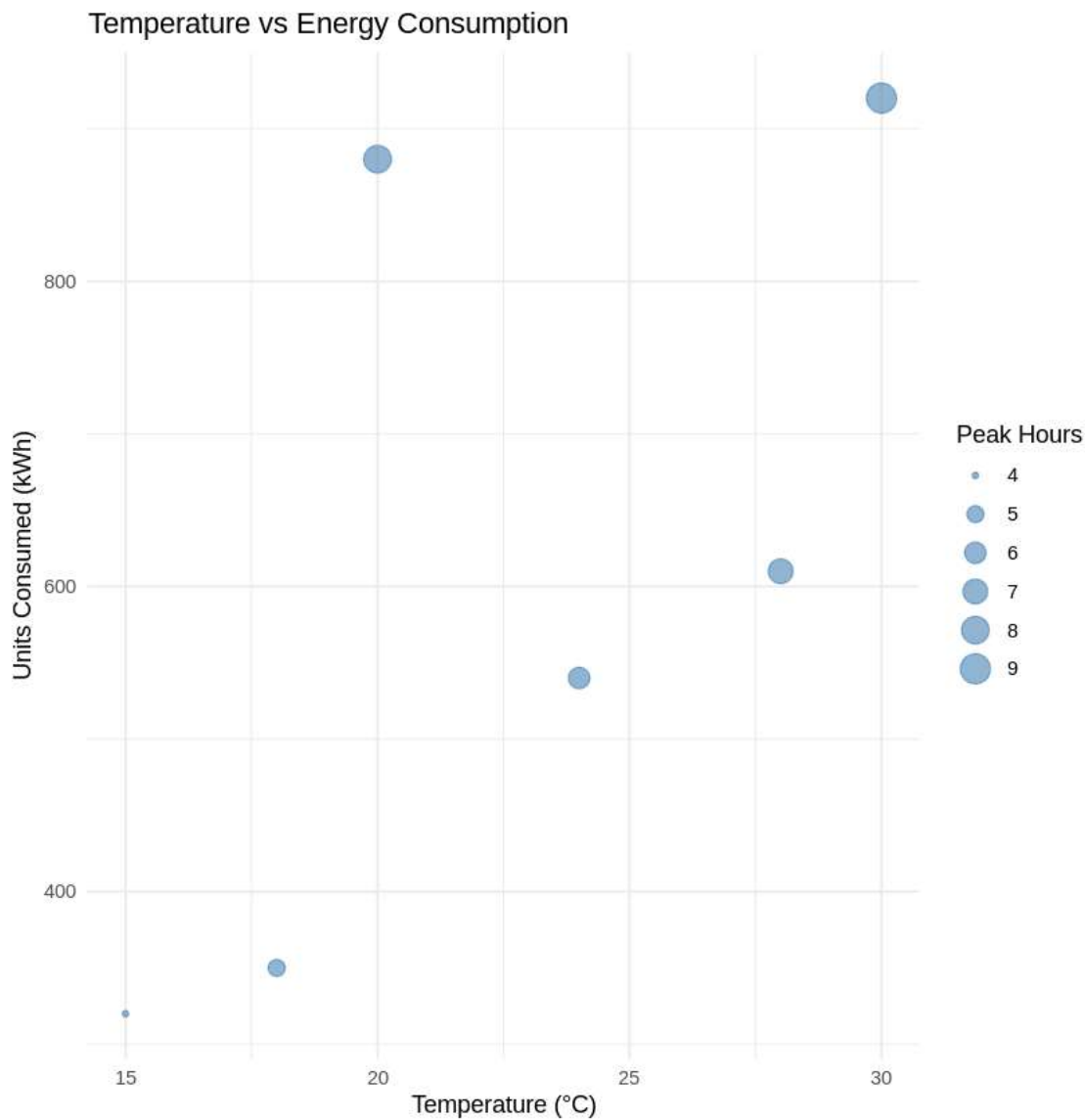
Using R, create a scatter plot between Temperature and Units_Consumed. Represent Peak_Hours using bubble size and apply transparency.

Code:

```
# Create the dataset
energy_data <- data.frame(
  Sector = c("Residential", "Commercial", "Industrial",
            "Residential", "Commercial", "Industrial"),
  Region = c("North", "South", "West", "East", "North", "South"),
  Month = c("Jan", "Jan", "Feb", "Feb", "Mar", "Mar"),
  Temperature = c(15, 24, 20, 18, 28, 30),
  Units_Consumed = c(320, 540, 880, 350, 610, 920),
  Cost = c(2100, 3600, 5900, 2300, 4100, 6200),
  Renewable_Usage = c(22, 18, 12, 25, 20, 15),
  Peak_Hours = c(4, 6, 8, 5, 7, 9)
)

library(ggplot2)
ggplot(
  energy_data,
  aes(
    x = Temperature,
    y = Units_Consumed,
    size = Peak_Hours
  )
) +
  geom_point(
    color = "steelblue",
    alpha = 0.6
  ) +
  labs(
    title = "Temperature vs Energy Consumption",
    x = "Temperature (°C)",
    y = "Units Consumed (kWh)",
    size = "Peak Hours"
  ) +
  theme_minimal()
```

Output:



SET 21 — Energy Consumption Analysis — Experiment 3

Using R, calculate average Renewable_Usage for each Sector. Create a bar chart and interpret renewable energy adoption.

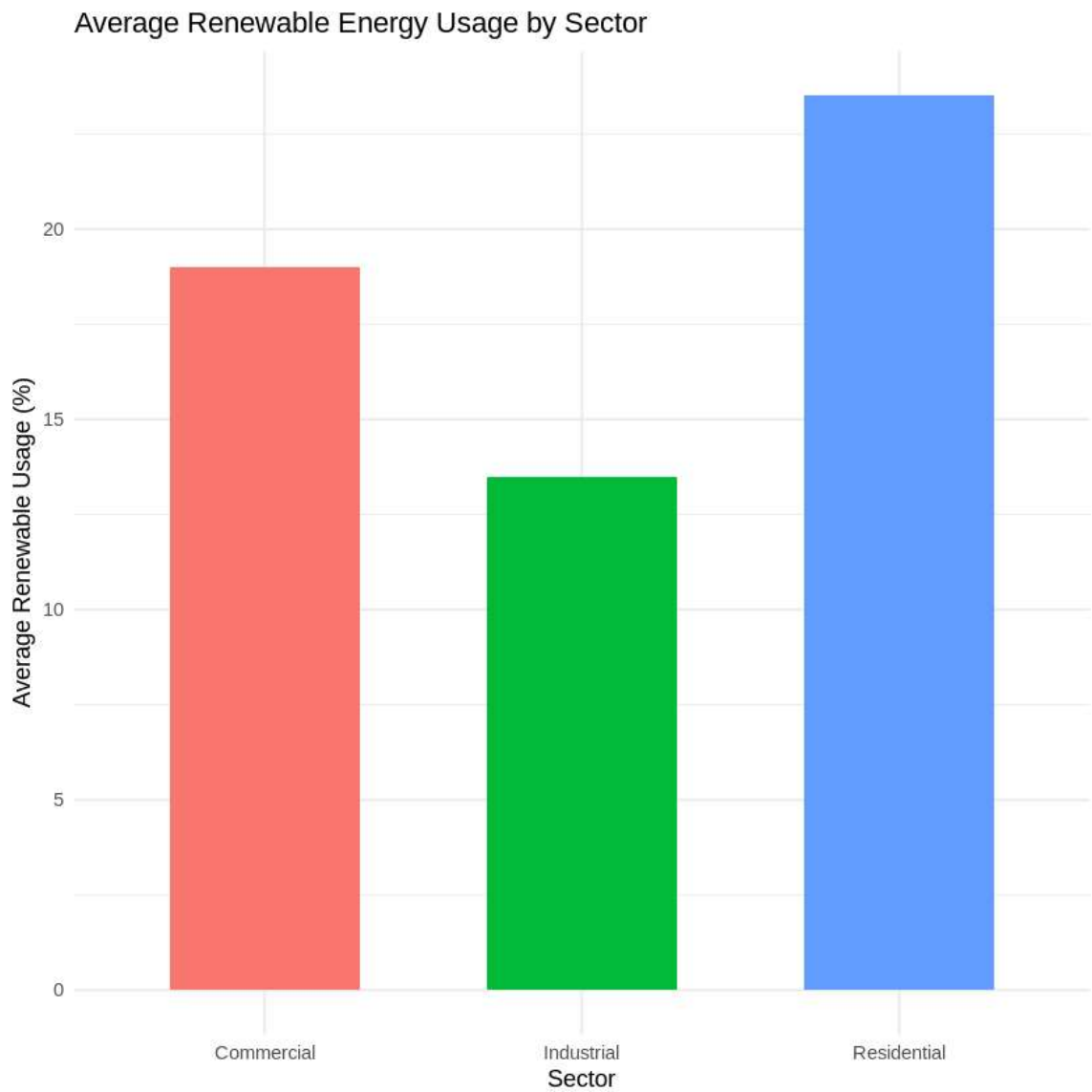
Code:

```
# Create the dataset
energy_data <- data.frame(
  Sector = c("Residential", "Commercial", "Industrial",
            "Residential", "Commercial", "Industrial"),
  Region = c("North", "South", "West", "East", "North", "South"),
  Month = c("Jan", "Jan", "Feb", "Feb", "Mar", "Mar"),
  Temperature = c(15, 24, 20, 18, 28, 30),
  Units_Consumed = c(320, 540, 880, 350, 610, 920),
  Cost = c(2100, 3600, 5900, 2300, 4100, 6200),
  Renewable_Usage = c(22, 18, 12, 25, 20, 15),
  Peak_Hours = c(4, 6, 8, 5, 7, 9)
)

library(ggplot2)
sector_average <- aggregate(
  Renewable_Usage ~ Sector,
  data = energy_data,
  FUN = mean
)

print(sector_average)
ggplot(
  sector_average,
  aes(
    x = Sector,
    y = Renewable_Usage,
    fill = Sector
  )
) +
  geom_bar(
    stat = "identity",
    width = 0.6
  ) +
  labs(
    title = "Average Renewable Energy Usage by Sector",
    x = "Sector",
    y = "Average Renewable Usage (%)"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```

Output:



SET 21 — Energy Consumption Analysis — Experiment 4

Import Energy_Consumption_Data.csv into Tableau. Create a monthly energy-usage line chart, Sector vs Region heatmap, Month and Sector filters, and an interactive dashboard.

Code:

Data Source:

Energy_Consumption_Data.csv

Sheet 1:

Line Chart

- Columns → Month
- Rows → SUM(Units_Consumed)

Sheet 2:

Heatmap

- Rows → Sector
- Columns → Region
- Marks → Square
- Color → SUM(Units_Consumed)

Filters:

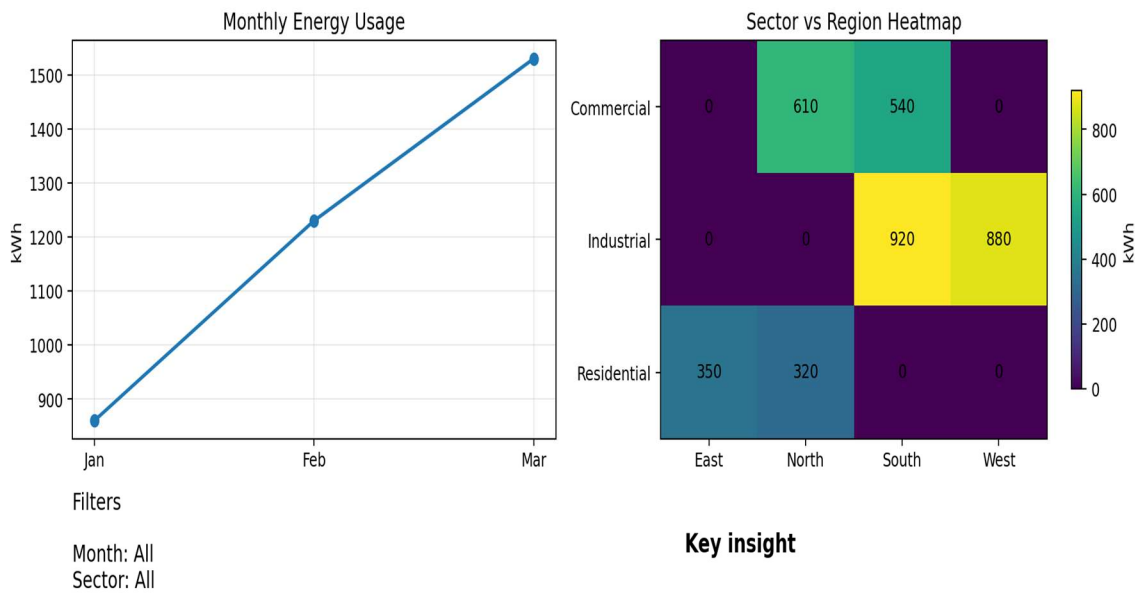
- Month
- Sector

Dashboard:

- Add the Line Chart
- Add the Heatmap
- Add Month filter
- Add Sector filter

Output:

Set 21 – Energy Consumption Dashboard



Key insight

Industrial usage is highest.
March has the highest total usage.

SET 22 — Time Series Analysis — Experiment 1

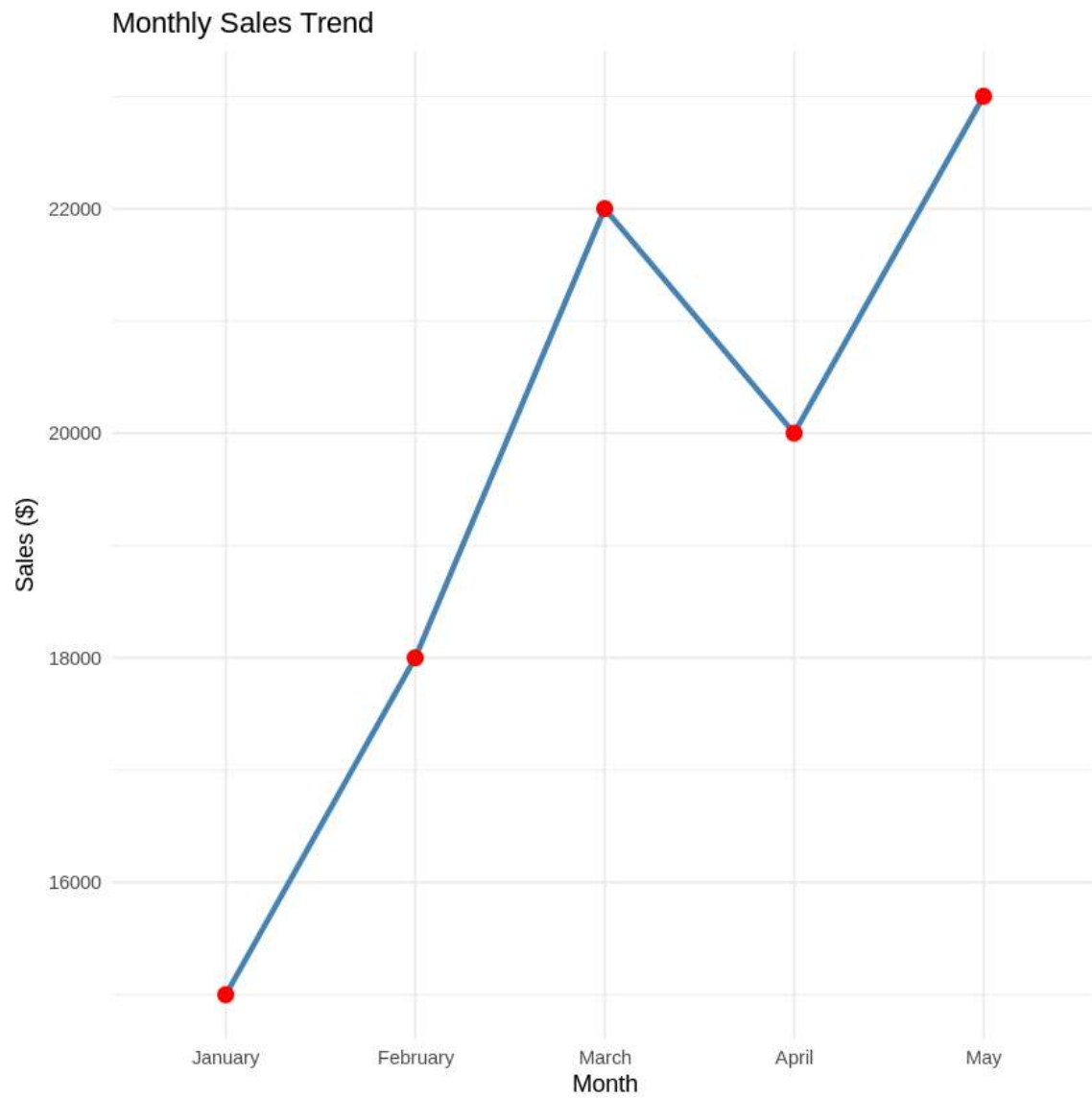
In R, create a time-series line chart to visualize the trend in monthly sales. Label axes and title.

Code:

```
sales_data <- data.frame(  
  Month = c(  
    "January",  
    "February",  
    "March",  
    "April",  
    "May"  
  ),  
  Sales = c(  
    15000,  
    18000,  
    22000,  
    20000,  
    23000  
  )  
)  
  
library(ggplot2)  
sales_data$Month <- factor(  
  sales_data$Month,  
  levels = c(  
    "January",  
    "February",  
    "March",  
    "April",  
    "May"  
  )  
)  
  
ggplot(  
  sales_data,  
  aes(  
    x = Month,  
    y = Sales,  
    group = 1  
  )  
) +  
  geom_line(  
    color = "steelblue",  
    linewidth = 1.2  
  ) +  
  geom_point(  
    color = "red",  
    size = 3  
  ) +  
  labs(  
    title = "Monthly Sales Trend",  
    x = "Month",  
    y = "Sales ($)"  
  )
```

```
) +  
theme_minimal()
```

Output:



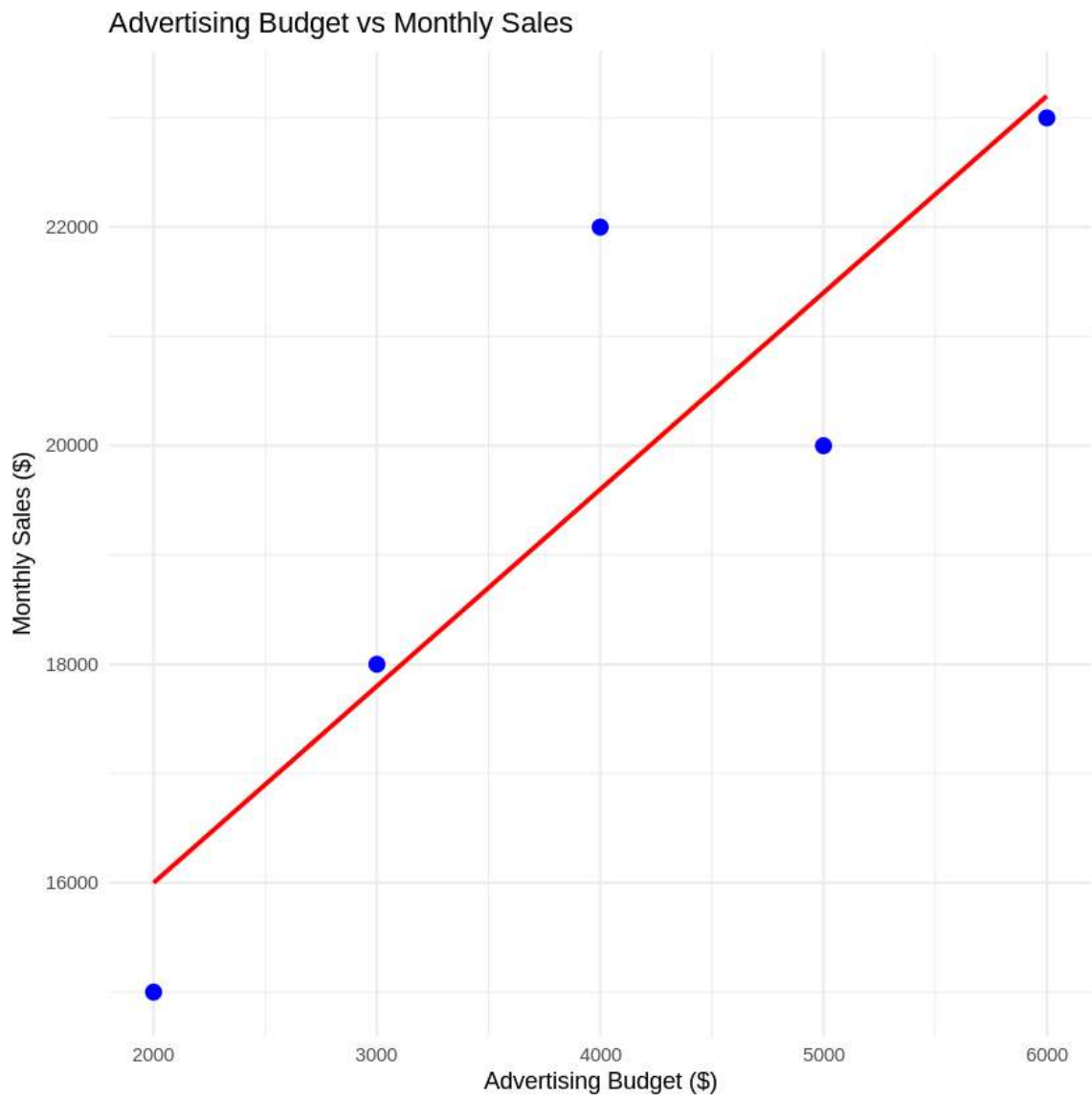
SET 22 — Time Series Analysis — Experiment 2

In R, generate a scatter plot to analyze the relationship between advertising budget and monthly sales. Explain insights.

Code:

```
advertising_data <- data.frame(  
  Advertising_Budget = c(  
    2000, 3000, 4000, 5000, 6000  
  ),  
  Sales = c(  
    15000, 18000, 22000, 20000, 23000  
  )  
)  
  
print(advertising_data)  
  
ggplot2::ggplot(  
  advertising_data,  
  ggplot2::aes(  
    x = Advertising_Budget,  
    y = Sales  
  )  
) +  
  ggplot2::geom_point(  
    color = "blue",  
    size = 3  
  ) +  
  ggplot2::geom_smooth(  
    method = "lm",  
    se = FALSE,  
    color = "red"  
  ) +  
  ggplot2::labs(  
    title = "Advertising Budget vs Monthly Sales",  
    x = "Advertising Budget ($)",  
    y = "Monthly Sales ($)"  
  ) +  
  ggplot2::theme_minimal()
```

Output:



SET 22 — Time Series Analysis — Experiment 3

In Tableau, build a dashboard combining a sales-trend line chart and a pie chart displaying sales distribution across products.

Code:

Sheet 1:

- Columns → Month
- Rows → Sales
- Marks → Line

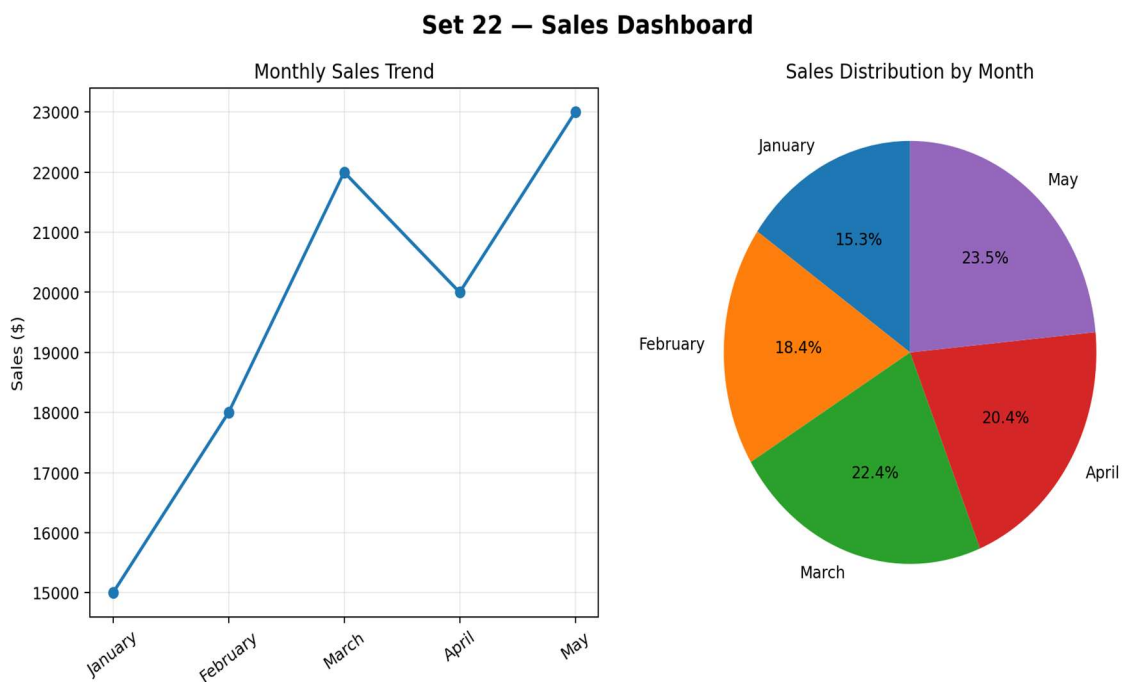
Sheet 2:

- Create a Pie Chart
- Use Product as the category
- Use Sales as the measure

Dashboard:

- Add Monthly Sales Line Chart
- Add Sales Distribution Pie Chart
- Arrange both charts on one dashboard

Output:



No Product field is supplied, so months are used for the second view.

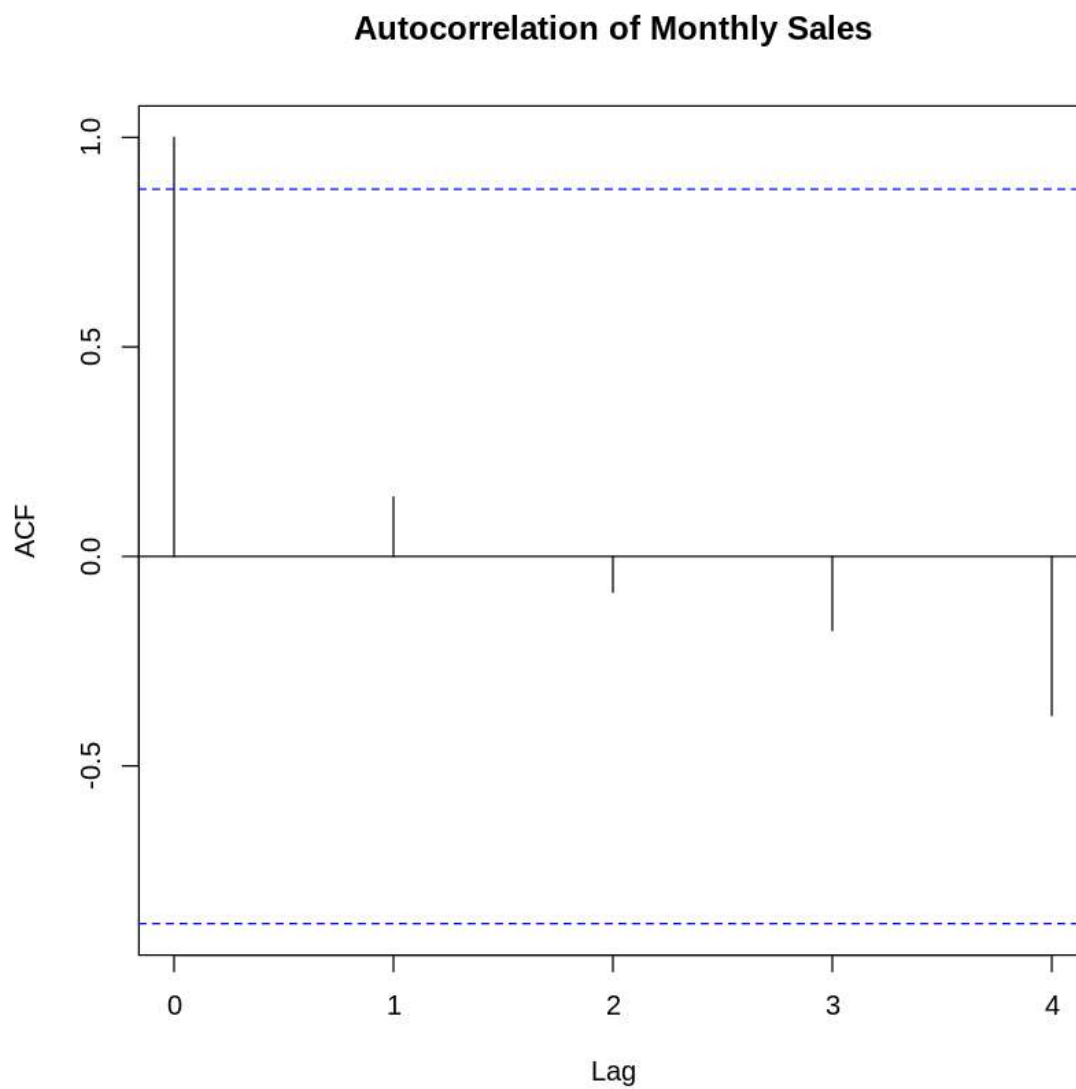
SET 22 — Time Series Analysis — Experiment 4

In R, create an autocorrelation plot to identify seasonality in the time-series data.

Code:

```
sales<-  
data.frame(Month=factor(c("January","February","March","April","May"),levels=c(  
"January","February","March","April","May"),ordered=TRUE),Sales=c(15000,18000,2  
2000,20000,23000))  
sales_ts<-ts(sales$Sales,frequency=12)  
acf(sales_ts,main="ACF of Monthly Sales")
```

Output:



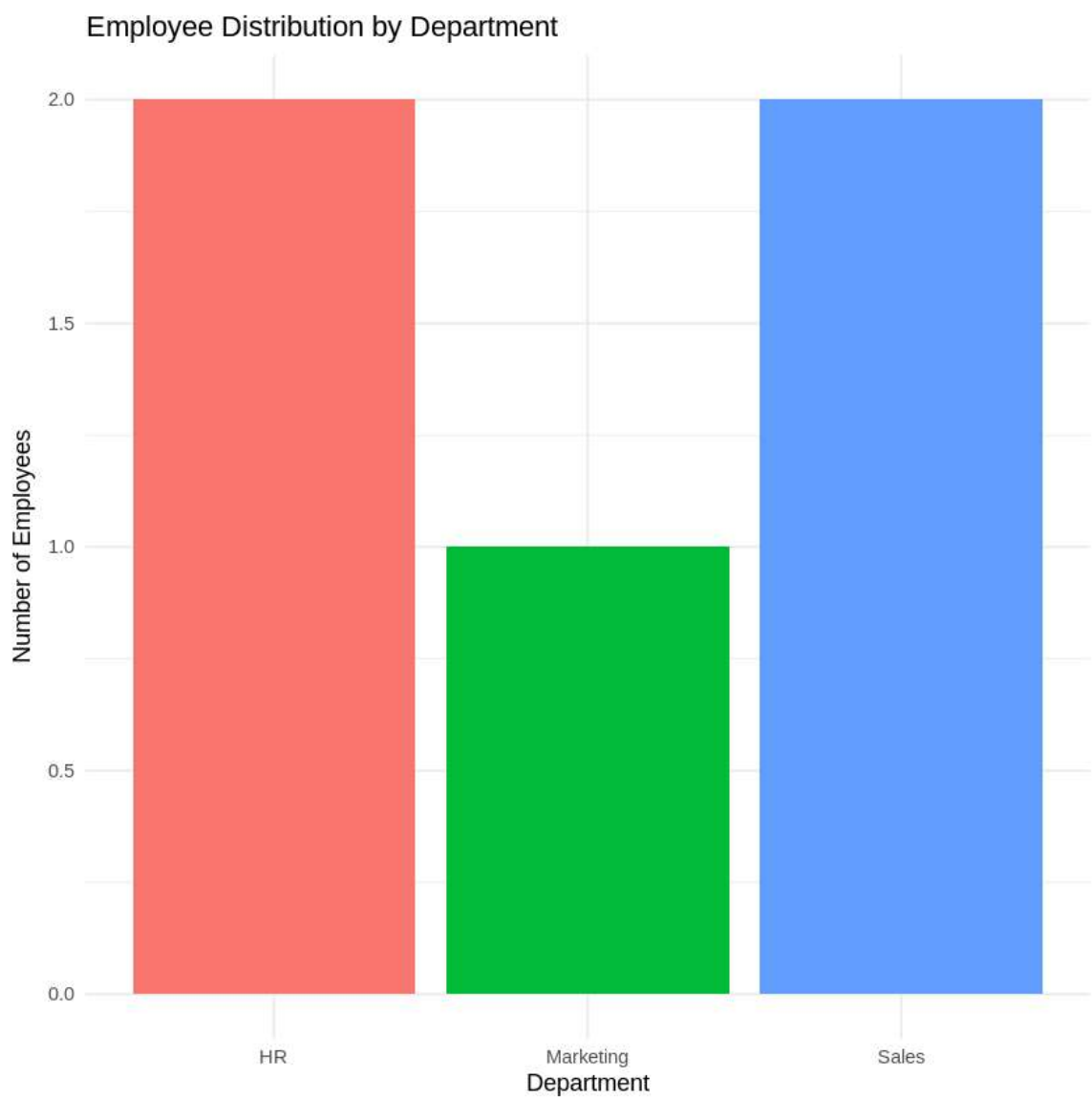
SET 23 — Employee Performance Analysis — Experiment 1

In R, create a bar chart showing the distribution of employees across departments. Label chart elements.

Code:

```
employee_data <- data.frame(  
  Employee_ID = c(1, 2, 3, 4, 5),  
  Department = c(  
    "Sales",  
    "HR",  
    "Marketing",  
    "Sales",  
    "HR"  
  ),  
  Years_of_Service = c(5, 3, 7, 4, 2),  
  Performance_Score = c(  
    85,  
    92,  
    78,  
    90,  
    76  
  )  
)  
  
print(employee_data)  
  
library(ggplot2)  
department_count <- as.data.frame(  
  table(employee_data$Department)  
)  
  
colnames(department_count) <- c(  
  "Department",  
  "Employee_Count"  
)  
  
ggplot(  
  department_count,  
  aes(  
    x = Department,  
    y = Employee_Count,  
    fill = Department  
  )  
) +  
  geom_bar(  
    stat = "identity"  
  ) +  
  labs(  
    title = "Employee Distribution by Department",  
    x = "Department",  
    y = "Number of Employees"  
  ) +  
  theme_minimal() +  
  theme(legend.position = "none")
```

Output:



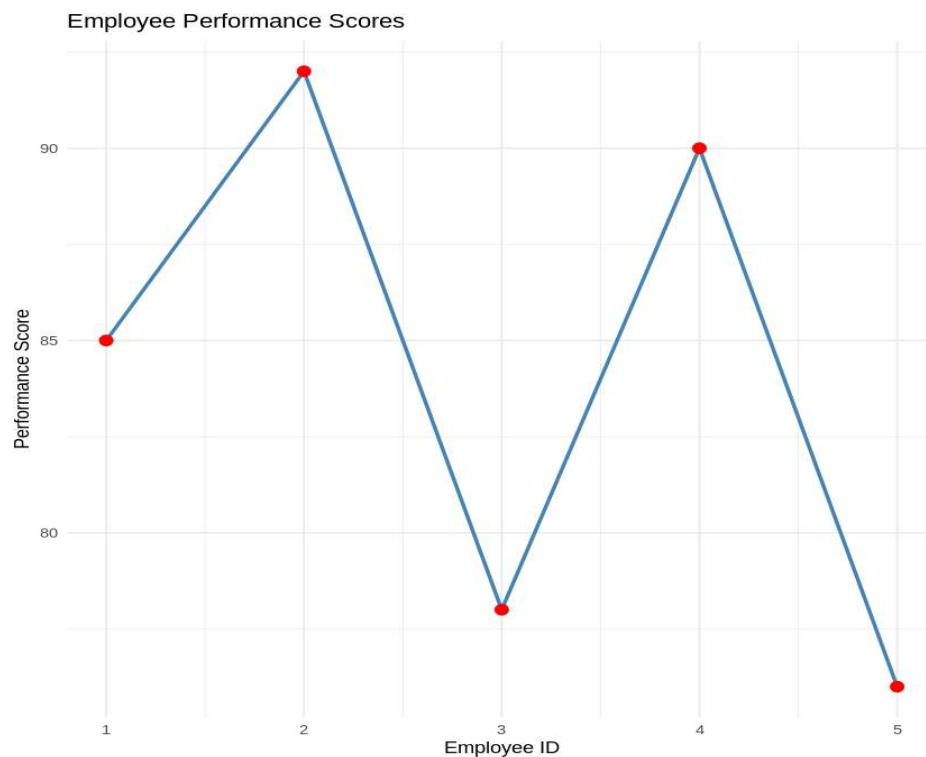
SET 23 — Employee Performance Analysis — Experiment 2

In R, generate a line chart to visualize the performance trend of employees over time.

Code:

```
ggplot(  
  employee_data,  
  aes(  
    x = Employee_ID,  
    y = Performance_Score  
  )  
) +  
  geom_line(  
    color = "steelblue",  
    linewidth = 1  
  ) +  
  geom_point(  
    color = "red",  
    size = 3  
  ) +  
  labs(  
    title = "Employee Performance Scores",  
    x = "Employee ID",  
    y = "Performance Score"  
  ) +  
  theme_minimal()
```

Output:



SET 23 — Employee Performance Analysis — Experiment 3

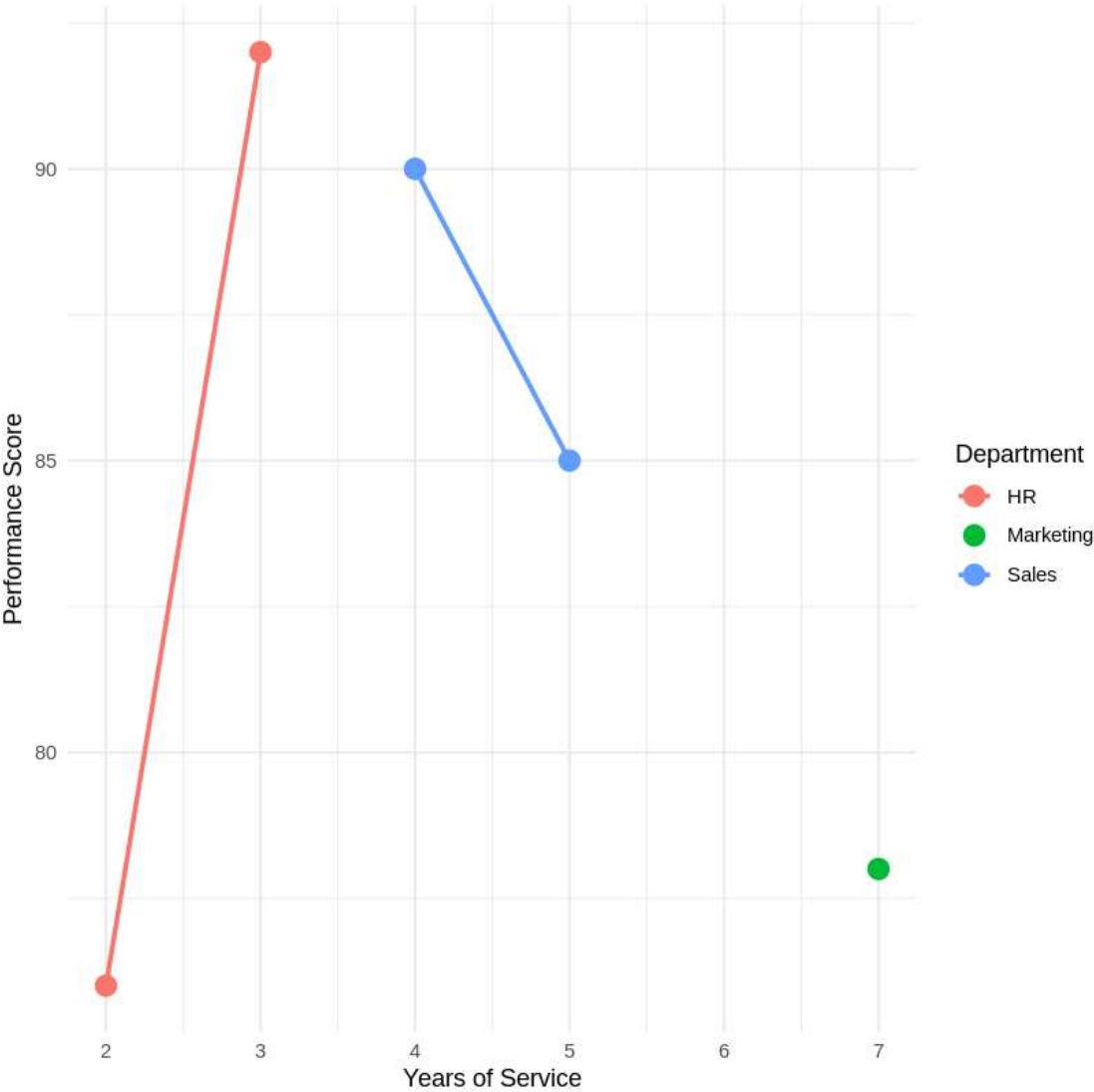
In Tableau, build a dashboard combining a department-distribution bar chart and a scatter plot of Years of Service vs Performance Score.

Code:

```
ggplot(  
  employee_data,  
  aes(  
    x = Years_of_Service,  
    y = Performance_Score,  
    color = Department  
  )  
) +  
  geom_point(  
    size = 4  
  ) +  
  geom_smooth(  
    method = "lm",  
    se = FALSE  
  ) +  
  labs(  
    title = "Years of Service vs Performance Score",  
    x = "Years of Service",  
    y = "Performance Score"  
  ) +  
  theme_minimal()
```

Output:

Years of Service vs Performance Score



SET 23 — Employee Performance Analysis — Experiment 4

In R, develop a table showing performance data for each employee. Label table elements.

Code:

```
library(knitr)

kable(
  employee_data,
  caption = "Employee Performance Data"
)
```

Output:

Table: Employee Performance Data

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78
4	Sales	4	90
5	HR	2	76

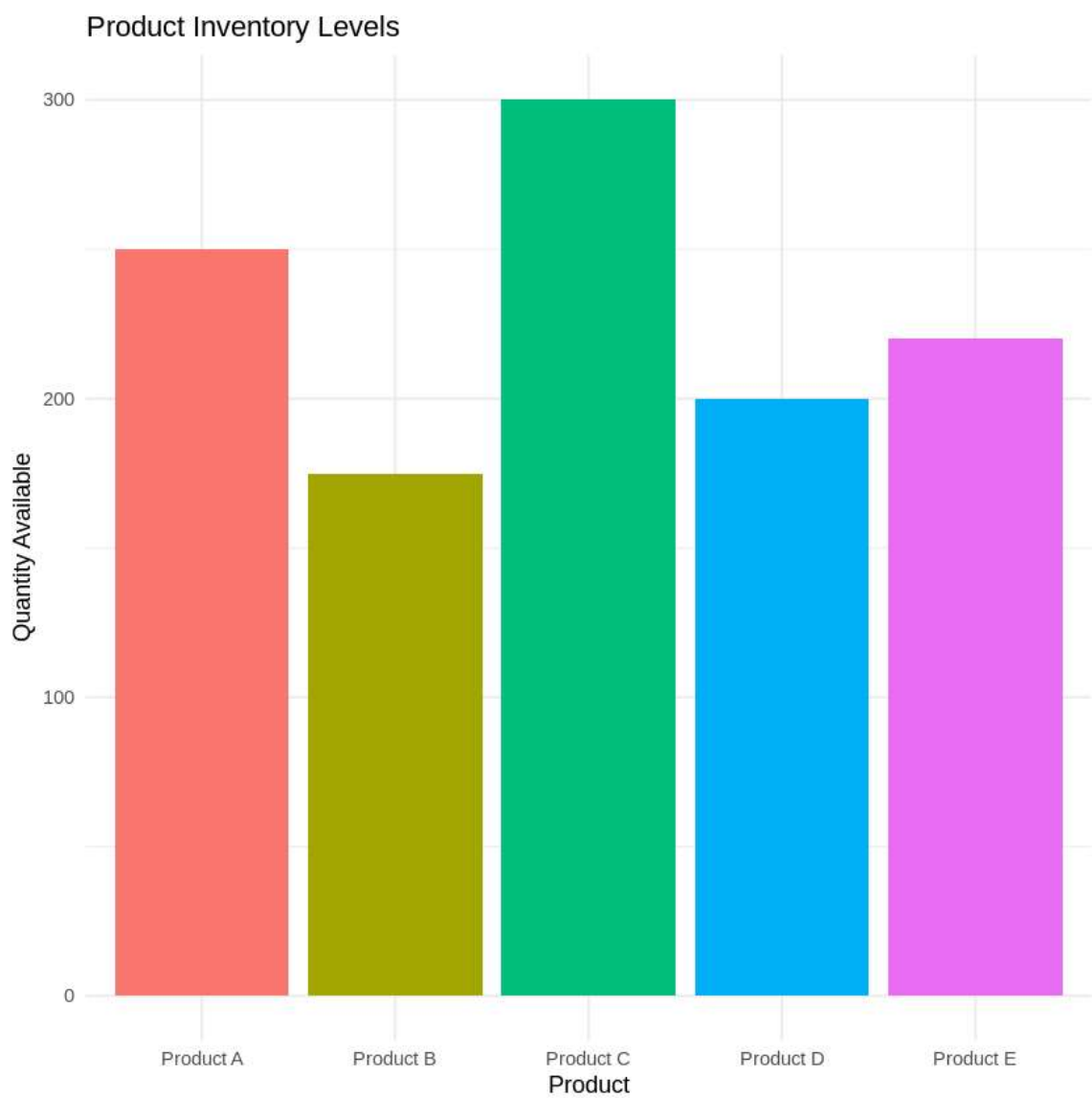
SET 24 — Product Inventory Management — Experiment 1

In R, create a bar chart showing quantity of each product. Label axes and title.

Code:

```
inventory_data <- data.frame(  
  Product_ID = c(1, 2, 3, 4, 5),  
  Product_Name = c(  
    "Product A",  
    "Product B",  
    "Product C",  
    "Product D",  
    "Product E"  
  ),  
  Quantity_Available = c(  
    250,  
    175,  
    300,  
    200,  
    220  
  )  
)  
  
library(ggplot2)  
ggplot(  
  inventory_data,  
  aes(  
    x = Product_Name,  
    y = Quantity_Available,  
    fill = Product_Name  
  )  
) +  
  geom_bar(  
    stat = "identity"  
  ) +  
  labs(  
    title = "Product Inventory Levels",  
    x = "Product",  
    y = "Quantity Available"  
  ) +  
  theme_minimal() +  
  theme(legend.position = "none")
```

Output:



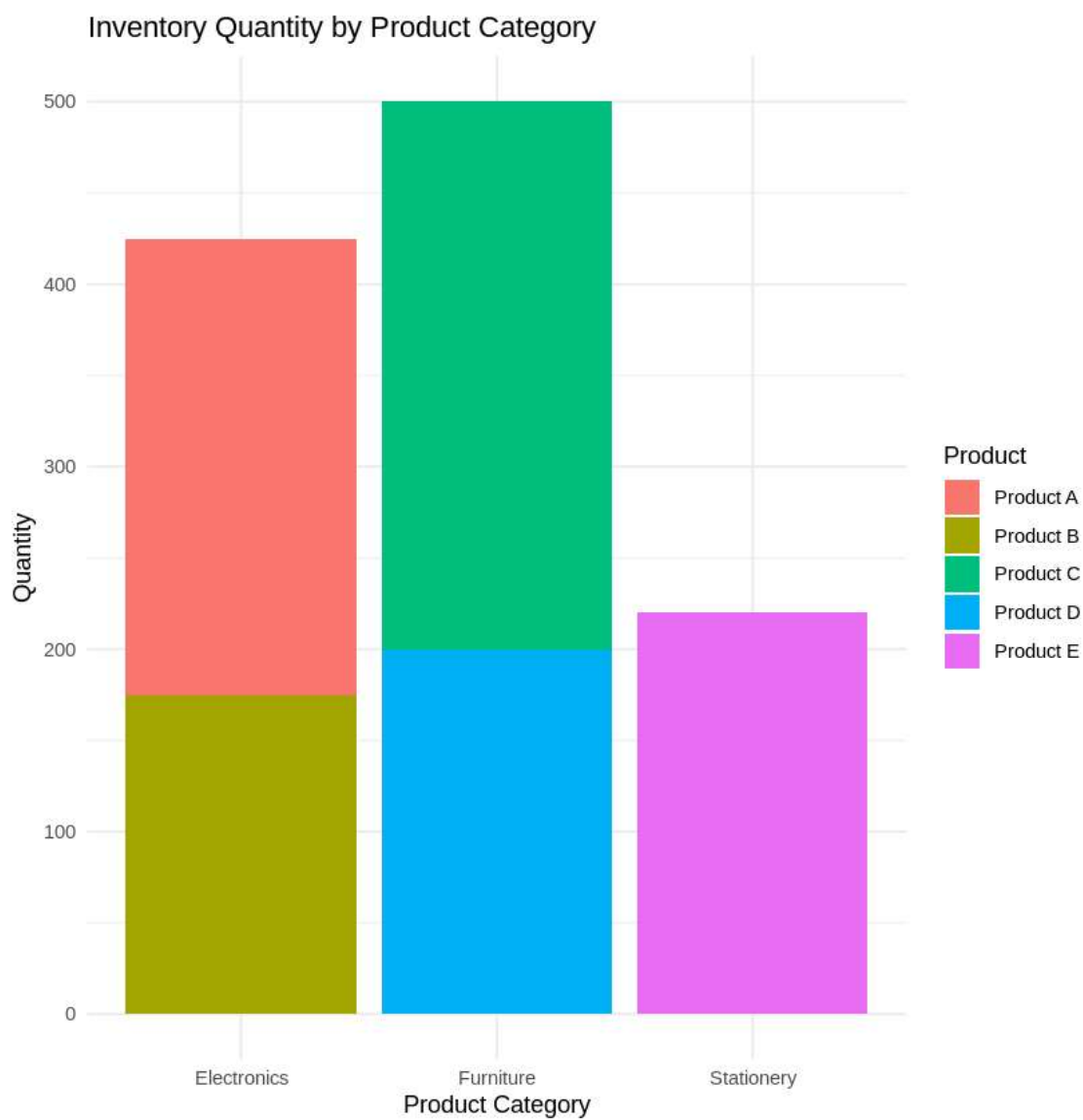
SET 24 — Product Inventory Management — Experiment 2

In R, generate a stacked bar chart to show quantity of each product within different product categories.

Code:

```
inventory_category <- data.frame(  
  Product = c(  
    "Product A",  
    "Product B",  
    "Product C",  
    "Product D",  
    "Product E"  
  ),  
  Category = c(  
    "Electronics",  
    "Electronics",  
    "Furniture",  
    "Furniture",  
    "Stationery"  
  ),  
  Quantity = c(  
    250,  
    175,  
    300,  
    200,  
    220  
  )  
)  
  
library(ggplot2)  
  
ggplot(  
  inventory_category,  
  aes(  
    x = Category,  
    y = Quantity,  
    fill = Product  
  )  
) +  
  geom_bar(  
    stat = "identity"  
  ) +  
  labs(  
    title = "Inventory Quantity by Product Category",  
    x = "Product Category",  
    y = "Quantity"  
  ) +  
  theme_minimal()
```

Output:



SET 24 — Product Inventory Management — Experiment 3

In Tableau, build a dashboard combining the bar chart and stacked bar chart for interactive inventory exploration.

Code:

Sheet 1:

- Product Name → Columns
- Quantity Available → Rows
- Marks → Bar

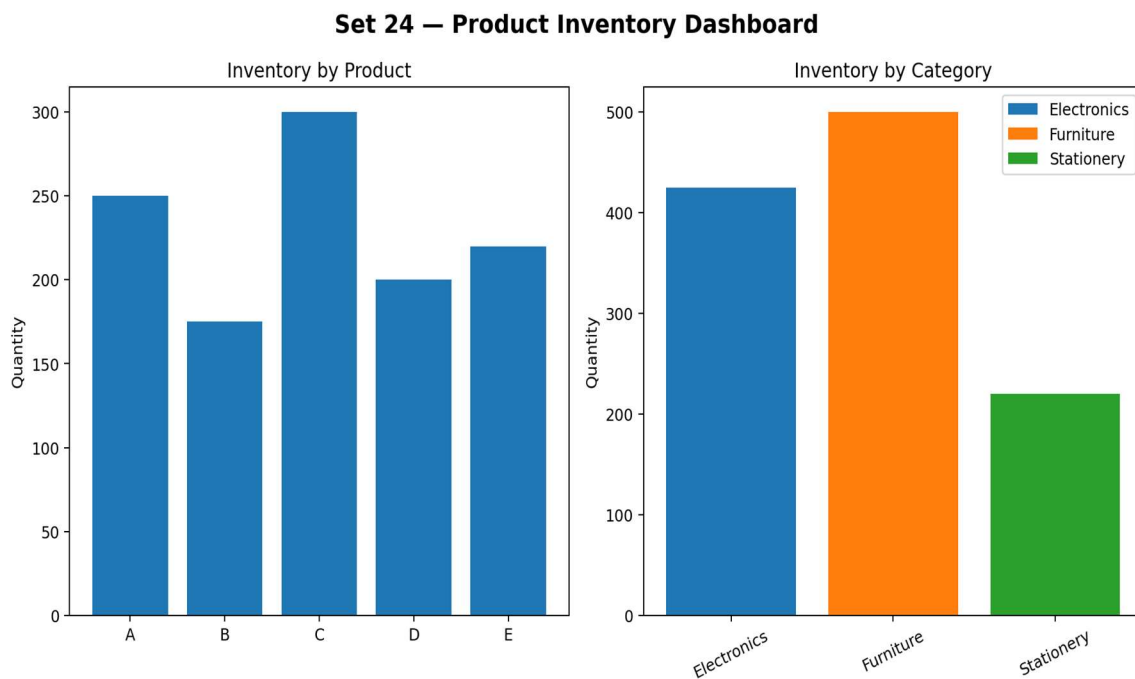
Sheet 2:

- Product Category → Columns
- Quantity → Rows
- Product → Color

Dashboard:

- Add both worksheets
- Add Product/Category filters if available
- Arrange the charts for interactive inventory exploration

Output:



Illustrative categories added because the supplied dataset has no Product Category field.

SET 24 — Product Inventory Management — Experiment 4

In R, create a scatter plot to explore Product Price vs Quantity Available and explain the relationship.

Code:

```
price_data <- data.frame(  
  Product = c(  
    "Product A",  
    "Product B",  
    "Product C",  
    "Product D",  
    "Product E"  
  ),  
  Price = c(  
    1200,  
    950,  
    1800,  
    1500,  
    700  
  ),  
  Quantity = c(  
    250,  
    175,  
    300,  
    200,  
    220  
  )  
)  
  
library(ggplot2)  
  
ggplot(  
  price_data,  
  aes(  
    x = Price,  
    y = Quantity  
  )  
) +  
  geom_point(  
    size = 4,  
    color = "steelblue"  
  ) +  
  labs(  
    title = "Product Price vs Quantity Available",  
    x = "Product Price",  
    y = "Quantity Available"  
  ) +  
  theme_minimal()
```

Output:



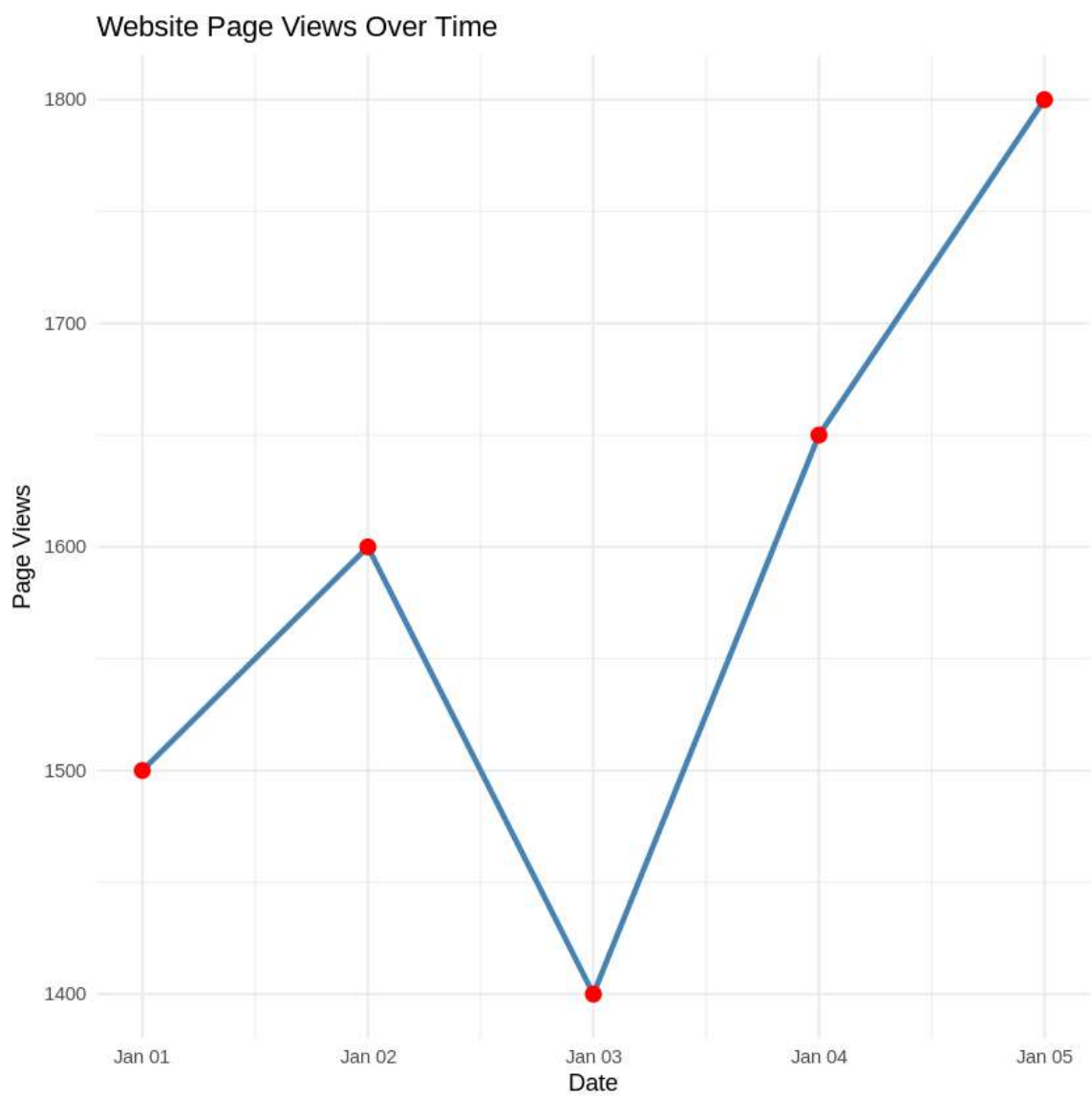
SET 25 — Website Traffic Analysis — Experiment 1

In R, create a line chart to visualize the trend in page views over time.

Code:

```
traffic_data <- data.frame(  
  Date = as.Date(c(  
    "2023-01-01",  
    "2023-01-02",  
    "2023-01-03",  
    "2023-01-04",  
    "2023-01-05"  
  )),  
  Page_Views = c(  
    1500,  
    1600,  
    1400,  
    1650,  
    1800  
  ),  
  Click_Through_Rate = c(  
    2.3,  
    2.7,  
    2.0,  
    2.4,  
    2.6  
  )  
)  
  
library(ggplot2)  
  
ggplot(  
  traffic_data,  
  aes(  
    x = Date,  
    y = Page_Views  
  )  
) +  
  geom_line(  
    color = "steelblue",  
    linewidth = 1.2  
  ) +  
  geom_point(  
    color = "red",  
    size = 3  
  ) +  
  labs(  
    title = "Website Page Views Over Time",  
    x = "Date",  
    y = "Page Views"  
  ) +  
  theme_minimal()
```

Output:



SET 25 — Website Traffic Analysis — Experiment 2

In R, generate a bar chart showing the top N days with the highest click-through rates. Label chart elements.

Code:

```
library(ggplot2)

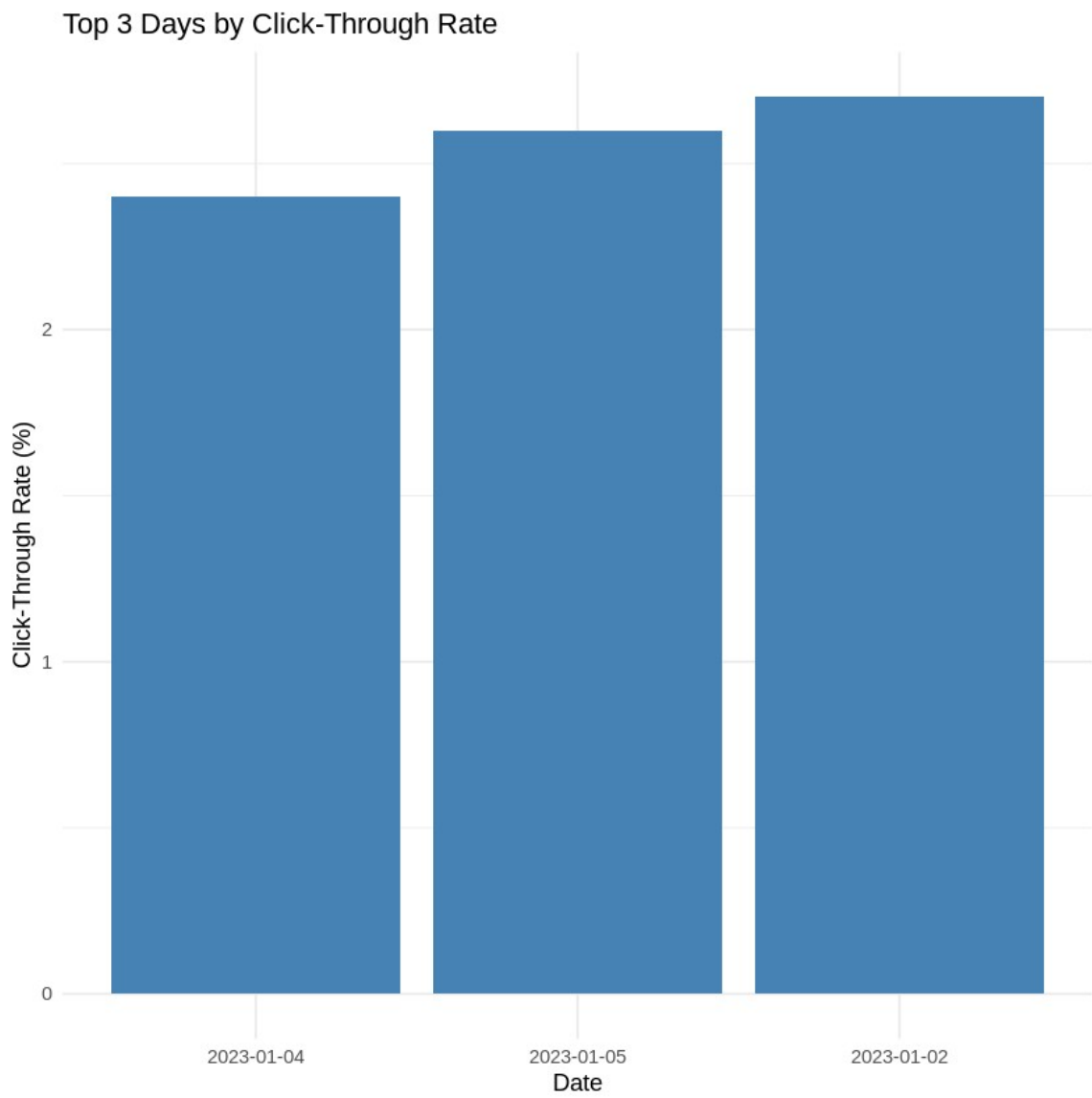
top_days <- traffic_data[
  order(
    traffic_data$Click_Through_Rate,
    decreasing = TRUE
  ),
]

top_days <- head(top_days, 3)

print(top_days)

ggplot(
  top_days,
  aes(
    x = reorder(
      as.character(Date),
      Click_Through_Rate
    ),
    y = Click_Through_Rate
  )
) +
  geom_bar(
    stat = "identity",
    fill = "steelblue"
  ) +
  labs(
    title = "Top 3 Days by Click-Through Rate",
    x = "Date",
    y = "Click-Through Rate (%)"
  ) +
  theme_minimal()
```

Output:



SET 25 — Website Traffic Analysis — Experiment 3

In R, build a stacked area chart showing Likes, Shares and Comments on the website.

Code:

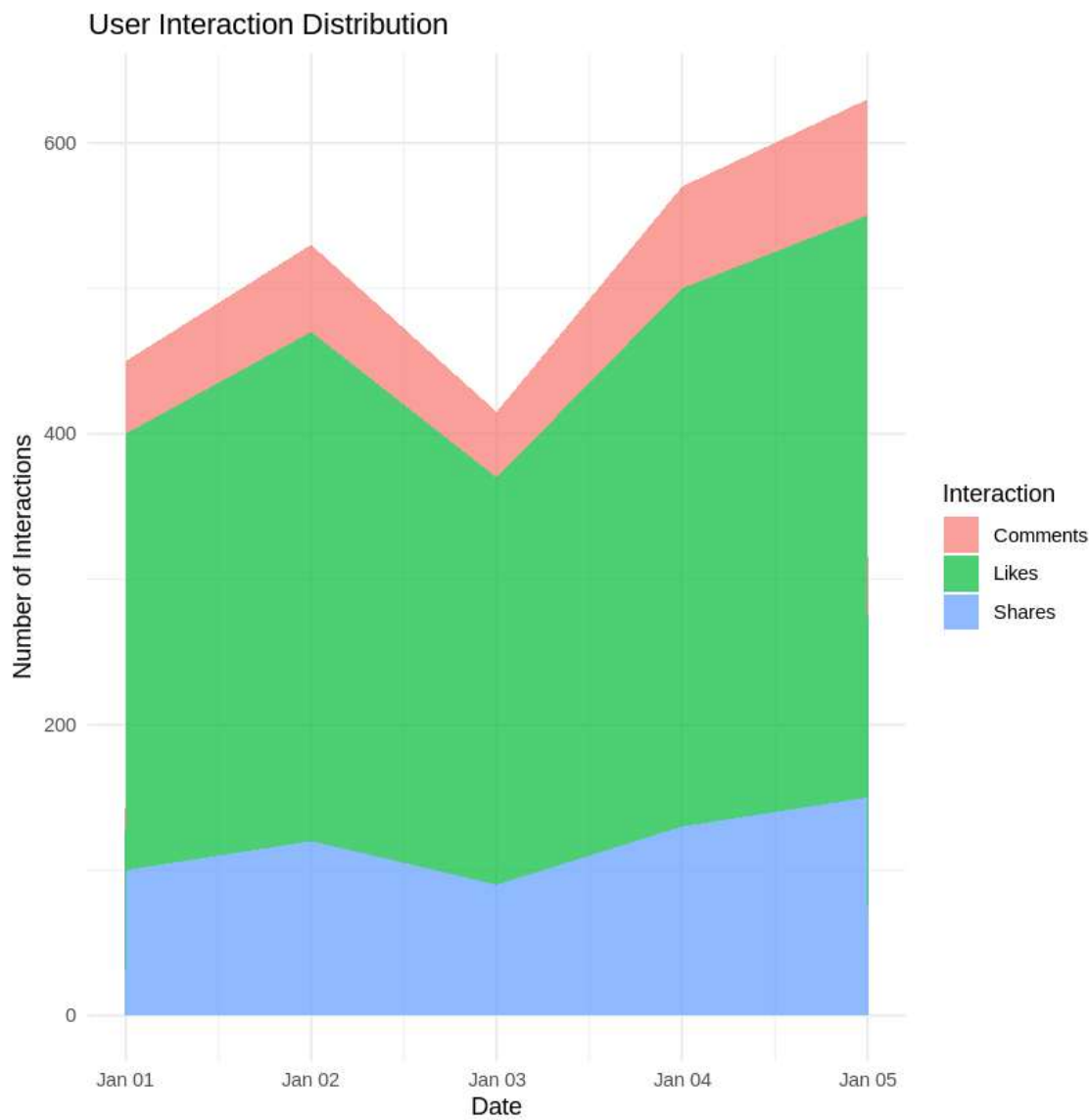
```
library(ggplot2)

interaction_data <- data.frame(
  Date = as.Date(c(
    "2023-01-01",
    "2023-01-02",
    "2023-01-03",
    "2023-01-04",
    "2023-01-05"
  )),
  Likes = c(300, 350, 280, 370, 400),
  Shares = c(100, 120, 90, 130, 150),
  Comments = c(50, 60, 45, 70, 80)
)

interaction_long <- data.frame(
  Date = rep(interaction_data$Date, 3),
  Interaction = rep(
    c("Likes", "Shares", "Comments"),
    each = 5
  ),
  Count = c(
    interaction_data$Likes,
    interaction_data$Shares,
    interaction_data$Comments
  )
)

ggplot(
  interaction_long,
  aes(
    x = Date,
    y = Count,
    fill = Interaction
  )
) +
  geom_area(alpha = 0.7) +
  labs(
    title = "User Interaction Distribution",
    x = "Date",
    y = "Number of Interactions"
  ) +
  theme_minimal()
```

Output:



SET 25 — Website Traffic Analysis — Experiment 4

In Tableau, create a dashboard with an interactive map showing traffic sources and a bar chart of page views by source.

Code:

Sheet 1:

- Date → Columns
- Page Views → Rows
- Marks → Line

Sheet 2:

- Traffic Source → Rows
- Page Views → Columns
- Marks → Bar

Dashboard:

- Add the line chart
- Add the source bar chart
- Add interactive filters
- Add map if geographic traffic-source data is available

Output:

